

2014-2015 Summer
MAT123 Final
23/08/2015

1. Sketch the graph of $f(x) = \ln(x^2 + 1)$.
2. Determine the area of the region bounded by $y = 4x + 3$ and $y = 6 - x - 2x^2$.
3. Use the Washer Method to find the volume of the solid obtained by revolving the region bounded by $y = 2\sqrt{x-1}$, $y = x - 1$ about the line $x = -1$, respectively.
4. Use the Cylindrical Shell Method to find the volume of the solid obtained by revolving the region bounded by $x = y^2 - 4$ and $x = 6 - 3y$ about $y = -8$.
5. Evaluate the following integrals.

a. $\int 4 \left(\frac{1}{x} - e^{-x} \right) \cos(e^{-x} + \ln x) \, dx$

b. $\int \frac{x^3 + 10x^2 + 3x + 36}{(x-1)^2 (x^2 + 4)^2} \, dx$

c. $\int \frac{\sqrt{25x^2 - 4}}{x} \, dx$

d. $\int x^2 \cos(4x) \, dx$

e. $\int \frac{dx}{2x^2 - 3x + 2}$

6. Investigate the convergence of the improper integral $\int_{-\infty}^0 (1 + 2x) e^{-x} \, dx$.
7. Evaluate the arc length $x = \frac{2}{3}(y-1)^{3/2}$ for $1 \leq y \leq 2$.

1. $\ln x$ is defined for $x > 0$. Therefore, $\ln(x^2 + 1)$ is defined on $\mathbb{R}$ because $x^2 + 1 \geq 1 > 0$.

Let us find the limit at infinity and negative infinity.

$$\lim_{x \rightarrow \infty} \ln(x^2 + 1) = \lim_{x \rightarrow -\infty} \ln(x^2 + 1) = \infty$$

No asymptotes occur.

Take the first derivative and find the critical points. Apply the chain rule.

$$y' = \frac{d}{dx} \ln(x^2 + 1) = \frac{1}{x^2 + 1} \cdot 2x = \frac{2x}{x^2 + 1}$$

A critical point occurs at $x = 0$. At this point, the first derivative is 0.

Take the second derivative. Apply the quotient rule.

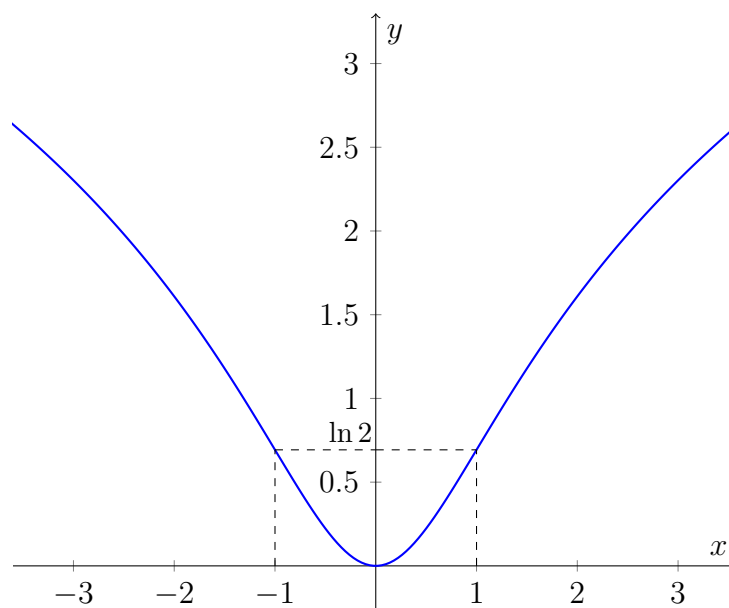
$$y'' = \frac{d}{dx} \left(\frac{2x}{x^2 + 1} \right) = \frac{2 \cdot (x^2 + 1) - 2x \cdot (2x)}{(x^2 + 1)^2} = \frac{2 - 2x^2}{(x^2 + 1)^2}$$

The inflection points occur at $x = \pm 1$. At these points, the direction of the curvature changes.

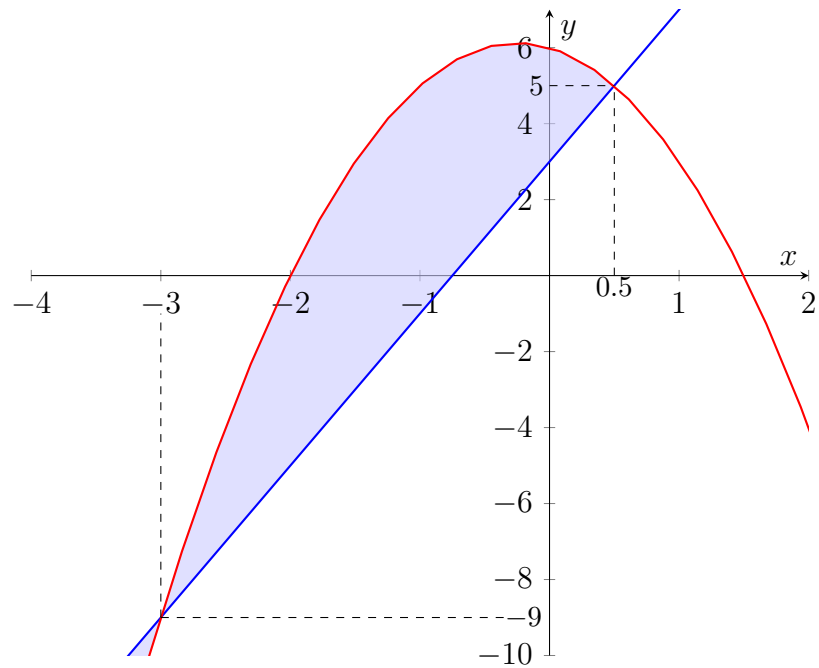
Consider some values of the function. Eventually, set up a table and see what the graph looks like in certain intervals.

$$f(-1) = f(1) = \ln 2, \quad f(0) = 0$$

x	$(-\infty, -1)$	$(-1, 0)$	$(0, 1)$	$(1, \infty)$
y	$(\ln 2, \infty)$	$(0, \ln 2)$	$(0, \ln 2)$	$(\ln 2, \infty)$
y' sign	-	-	+	+
y'' sign	-	+	+	-

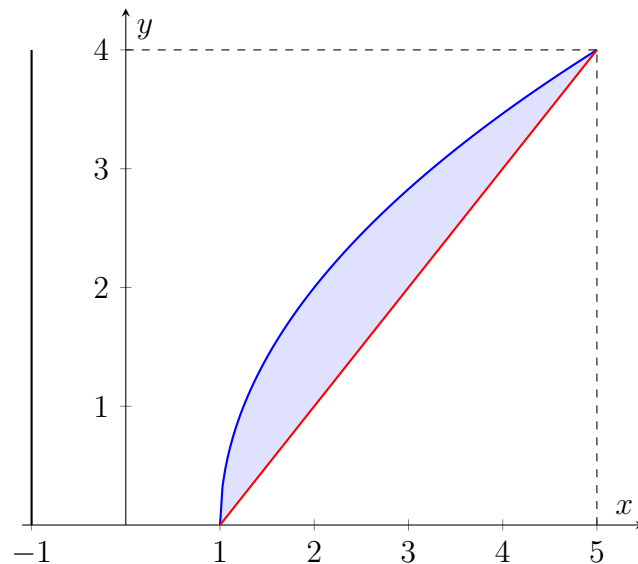


2.



$$\begin{aligned} \text{Area} &= \int_{-3}^{1/2} [6 - x - 2x^2 - (4x + 3)] \, dx = \int_{-3}^{1/2} (3 - 5x - 2x^2) \, dx \\ &= 3x - \frac{5x^2}{2} - \frac{2x^3}{3} \Big|_{-3}^{1/2} = \left(\frac{3}{2} - \frac{5}{8} - \frac{1}{12} \right) - \left(-9 - \frac{45}{2} + 18 \right) = \boxed{\frac{343}{24}} \end{aligned}$$

3.



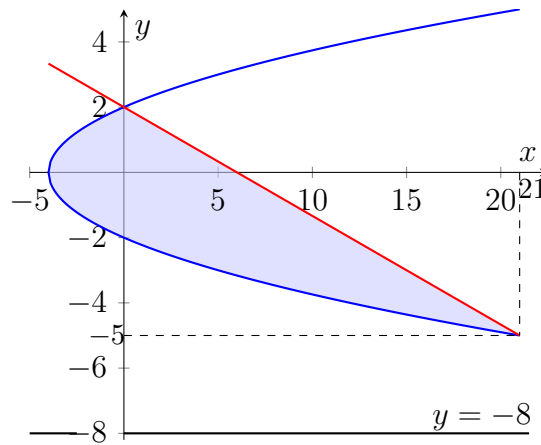
Rewrite the equations of the curves and solve for x . Let V be the volume of the solid.

$$y = 2\sqrt{x-1} \implies y^2 = 4x - 4 \implies x = \frac{y^2 + 4}{4}$$

$$y = x - 1 \implies x = y + 1$$

$$\begin{aligned}
V &= \int_D \pi [R_2^2(y) - R_1^2(y)] dy = \int_0^4 \pi \left[((y+1)+1)^2 - \left(\left(\frac{y^2+4}{4} \right) + 1 \right)^2 \right] dy \\
&= \pi \int_0^4 \left[(y+2)^2 - \left(\frac{y^2}{4} + 2 \right)^2 \right] dy = \pi \int_0^4 \left(y^2 + 4y + 4 - \frac{y^4}{16} - y^2 - 4 \right) dy \\
&= \pi \int_0^4 \left(4y - \frac{y^4}{16} \right) dy = \pi \left[2y^2 - \frac{y^5}{80} \right]_0^4 = \pi \left[\left(32 - \frac{64}{5} \right) - (0) \right] = \boxed{\frac{96\pi}{5}}
\end{aligned}$$

4.



Let V be the volume of the solid.

$$\begin{aligned}
V &= \int_D 2\pi \cdot h(y) \cdot r(y) dy = 2\pi \int_{-5}^2 (y+8) \cdot [(6-y^2) - (y^2-4)] dy \\
&= 2\pi \int_{-5}^2 (y+8)(-y^2-3y+10) dy \\
&= 2\pi \int_{-5}^2 (-y^3-3y^2+10y-8y^2-24y+80) dy \\
&= 2\pi \int_{-5}^2 (-y^3-11y^2-14y+80) dy = 2\pi \left[-\frac{y^4}{4} - \frac{11y^3}{3} - 7y^2 + 80y \right]_{-5}^2 \\
&= 2\pi \left[\left(-4 - \frac{88}{3} - 28 + 160 \right) - \left(\frac{625}{4} + \frac{1375}{3} - 175 - 400 \right) \right] = \boxed{\frac{4459\pi}{6}}
\end{aligned}$$

5. Evaluate the following integrals.

- a. Use the u -substitution method. Let $u = e^{-x} + \ln x$. Then $du = (-e^{-x} + \frac{1}{x}) dx$.

$$\begin{aligned} \int 4 \left(\frac{1}{x} - e^{-x} \right) \cos(e^{-x} + \ln x) dx &= \int 4 \cos u du = 4 \sin u + C \\ &= \boxed{4 \sin(e^{-x} + \ln x) + C, C \in \mathbb{R}} \end{aligned}$$

- b. Decompose the fraction into multiple partial fractions. Let $A, B, C, D, E, F \in \mathbb{R}$.

$$\begin{aligned} I &= \int \frac{x^3 + 10x^2 + 3x + 36}{(x-1)^2 (x^2 + 4)^2} dx \\ &= \int \left(\frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{Cx+D}{x^2+4} + \frac{Ex+F}{(x^2+4)^2} \right) dx \end{aligned}$$

Let $N = x^3 + 10x^2 + 3x + 36$ and $C_0, C_1, C_2, \dots, C_6$ be arbitrary constants.

$$\begin{aligned} N &= A(x^2 + 4)^2(x-1) + B(x^2 + 4)^2 + (Cx + D)(x^2 + 4)(x-1)^2 \\ &\quad + (Ex + F)(x-1)^2 \\ &= (x^2 + 4)^2 [A(x-1) + B] + (x-1)^2 [(Cx + D)(x^2 + 4) + Ex + F] \\ &= (x^4 + 8x^2 + 16)(Ax - A + B) + (x^2 - 2x + 1) \\ &\quad \cdot (Cx^3 + 4Cx + Dx^2 + 4D + Ex + F) \\ &= Ax^5 - Ax^4 + Bx^4 + 8Ax^3 - 8Ax^2 + 8Bx^2 + 16Ax - 16A + 16B \\ &\quad + Cx^5 + 4Cx^3 + Dx^4 + 4Dx^2 + Ex^3 + Fx^2 - 2Cx^4 - 8Cx^2 - 2Dx^3 \\ &\quad - 8Dx - 2Ex^2 - 2Fx + Cx^3 + 4Cx + Dx^2 + 4D + Ex + F \\ &= x^5(A + C) + x^4(-A + B + D - 2C) + x^3(8A + 4C + E - 2D + C) \\ &\quad + x^2(-8A + 8B + 4D + F - 8C - 2E + D) \\ &\quad + x(16A - 8D - 2F + 4C + E) - 16A + 16B + 4D + F \end{aligned}$$

Equate the coefficients of like terms.

$$\begin{aligned} A + C &= 0 \\ -A + B + D - 2C &= 0 \\ 8A + 5C + E - 2D &= 1 \\ -8A + 8B + 5D - 8C + F - 2E &= 10 \\ 16A - 8D - 2F + 4C + E &= 3 \\ -16A + 16B + 4D + F &= 36 \end{aligned} \tag{1}$$

From (1), $A = -C$. Rewrite C in terms of A and rearrange the equations.

$$\begin{aligned} A + B + D &= 0 & (2) \\ 3A + E - 2D &= 1 \\ 8B + 5D + F - 2E &= 10 \\ 12A - 8D - 2F + E &= 3 \\ -16A + 16B + 4D + F &= 36 \end{aligned}$$

From (2), $A + B = -D$. Rewrite D in terms of A and B and rearrange the equations.

$$\begin{aligned} 5A + 2B + E &= 1 & (3) \\ -5A + 3B + F - 2E &= 10 & (4) \\ 20A + 8B - 2F + E &= 3 & (5) \\ -20A + 12B + F &= 36 & (6) \end{aligned}$$

By using the couples (3) & (4) and (5) & (6), eliminate E .

$$\left. \begin{aligned} 5A + 7B + F &= 12 \\ 35A + 19B - 3F &= 16 \\ -20A + 12B + F &= 36 \end{aligned} \right\} \Rightarrow \left. \begin{aligned} 50A + 40B &= 52 \\ -25A + 55B &= 124 \end{aligned} \right\} \Rightarrow \begin{aligned} A &= -\frac{14}{25} \\ B &= 2 \end{aligned}$$

Therefore, $C = \frac{14}{25}$, $D = -\frac{36}{25}$. From (3), $E = -\frac{1}{5}$, and from (6), $F = \frac{4}{5}$.
Substitute the values into A , B , C , D , E , F .

$$I = \int \left(-\frac{14}{25} \cdot \frac{1}{x-1} + \frac{2}{(x-1)^2} + \frac{\frac{14}{25}x - \frac{36}{25}}{x^2 + 4} + \frac{-\frac{1}{5}x + \frac{4}{5}}{(x^2 + 4)^2} \right) dx \quad (7)$$

From now on, integrate term by term. Integrate the first term in (7).

$$\int -\frac{14}{25} \cdot \frac{1}{x-1} dx = -\frac{14}{25} \int \frac{1}{x-1} dx = -\frac{14}{25} \ln |x-1| + C_1 \quad (8)$$

Integrate the second term in (7).

$$\int \frac{2}{(x-1)^2} dx = -\frac{2}{x-1} + C_2 \quad (9)$$

Integrate the third term in (7).

$$\begin{aligned}
 \int \frac{\frac{14}{25}x - \frac{36}{25}}{x^2 + 4} dx &= \frac{1}{25} \int \frac{14x - 36}{x^2 + 4} dx = \frac{7}{25} \int \frac{2x}{x^2 + 4} dx - \frac{36}{25} \int \frac{1}{x^2 + 4} dx \\
 &= \frac{7}{25} \ln |x^2 + 4| - \frac{36}{100} \int \frac{1}{\left(\frac{x}{2}\right)^2 + 1} dx \\
 &= \frac{7}{25} \ln (x^2 + 4) - \frac{36}{50} \arctan \left(\frac{x}{2}\right) + C_3
 \end{aligned} \tag{10}$$

Integrate the last term in (7).

$$\int \frac{-\frac{1}{5}x + \frac{4}{5}}{(x^2 + 4)^2} dx = \frac{1}{5} \int \frac{4 - x}{(x^2 + 4)^2} dx = \frac{4}{5} \int \frac{1}{(x^2 + 4)^2} dx - \frac{1}{5} \int \frac{x}{(x^2 + 4)^2} dx \tag{11}$$

First, solve the integral on the left in (11). Let $x = 2 \tan u$, then $dx = 2 \sec^2 u \, du$.

$$\begin{aligned}
 \int \frac{1}{(x^2 + 4)^2} dx &= \int \frac{2 \sec^2 u}{(4 \tan^2 u + 4)^2} du = \int \frac{2 \sec^2 u}{16 \sec^4 u} du = \frac{1}{8} \int \frac{1}{\sec^2 u} du \\
 &= \frac{1}{8} \int \cos^2 u \, du = \frac{1}{8} \int \left(\frac{1 + \cos 2u}{2} \right) du = \frac{1}{8} \left(\frac{u}{2} + \frac{\sin 2u}{4} \right) + C_4 \\
 &= \frac{u}{16} + \frac{\sin u \cos u}{16} + C_4
 \end{aligned}$$

We have $x = 2 \tan u$, $\tan u = \frac{x}{2} \implies u = \arctan \frac{x}{2}$, $\sin u = \frac{x}{\sqrt{x^2 + 4}}$, $\cos u = \frac{2}{\sqrt{x^2 + 4}}$.

Rewrite the integral.

$$\int \frac{1}{(x^2 + 4)^2} dx = \frac{1}{16} \left(\arctan \frac{x}{2} + \frac{2x}{x^2 + 4} \right) + C_4 \tag{12}$$

Now, solve the integral on the right in (11). Let $u = x^2 + 4$, then $du = 2x \, dx$.

$$\int \frac{x}{(x^2 + 4)^2} dx = \int \frac{du}{2u^2} = -\frac{1}{2u} + C_5 = -\frac{1}{2(x^2 + 4)} + C_5 \tag{13}$$

Rewrite the integral in (11) using (12) and (13).

$$\frac{4}{5} \int \frac{1}{(x^2 + 4)^2} dx - \frac{1}{5} \int \frac{x}{(x^2 + 4)^2} dx = \frac{1}{20} \left(\arctan \frac{x}{2} + \frac{2x}{x^2 + 4} \right) + C_6 \tag{14}$$

Eventually, using (8), (9), (10), and (14), rewrite (7).

$$\boxed{I = -\frac{14}{25} \ln |x - 1| - \frac{2}{x - 1} + \frac{7}{25} \ln (x^2 + 4) - \frac{67}{100} \arctan \frac{x}{2} + \frac{x + 1}{10(x^2 + 4)} + C_0}$$

c. Let $x = \frac{2}{5} \sec u$ for $0 \leq u < \frac{\pi}{2}$, then $dx = \frac{2}{5} \sec u \tan u \, du$.

$$\begin{aligned} I &= \int \frac{\sqrt{25x^2 - 4}}{x} dx = \int \frac{\sqrt{4\sec^2 u - 4}}{\frac{2}{5}\sec u} \cdot \frac{2}{5} \sec u \tan u \, du \quad [\tan^2 u + 1 = \sec^2 u] \\ &= 2 \int |\tan u| \tan u \, du \quad [\tan u > 0] \\ &= 2 \int \tan^2 u \, du = 2 \int \sec^2 u \, du - 2 \int du = 2 \tan u - 2u + C \end{aligned}$$

Recall that $x = \frac{2}{5} \sec u$.

$$\begin{aligned} \sec u = \frac{5x}{2} &\implies \sec^2 u = \frac{25x^2}{4} \implies \tan u = \frac{\sqrt{25x^2 - 4}}{2} \\ &\implies u = \arctan\left(\frac{\sqrt{25x^2 - 4}}{2}\right) \end{aligned}$$

$$\int \frac{\sqrt{25x^2 - 4}}{x} dx = \boxed{\sqrt{25x^2 - 4} - 2 \arctan\left(\frac{\sqrt{25x^2 - 4}}{2}\right) + C, \quad C \in \mathbb{R}}$$

d. Use the method of integration by parts.

$$\left. \begin{aligned} u &= x^2 \implies du = 2x \, dx \\ dv &= \cos(4x) \, dx \implies v = \frac{1}{4} \sin(4x) \end{aligned} \right\} \rightarrow \int u \, dv = uv - \int v \, du$$

$$I = x^2 \cdot \frac{1}{4} \sin(4x) - \int \frac{1}{4} \sin(4x) \cdot 2x \, dx = \frac{x^2}{4} \sin(4x) - \frac{1}{2} \int x \sin(4x) \, dx$$

Apply the same procedure.

$$\left. \begin{aligned} w &= x \implies dw = dx \\ dz &= \sin(4x) \, dx \implies z = -\frac{1}{4} \cos(4x) \end{aligned} \right\} \rightarrow \int w \, dz = wz - \int z \, dw$$

$$I = \frac{x^2}{4} \sin(4x) - \frac{1}{2} \left[\frac{-x}{4} \cos(4x) - \int -\frac{1}{4} \cos(4x) \, dx \right]$$

$$= \boxed{\frac{x^2}{4} \sin(4x) + \frac{x}{8} \cos(4x) - \frac{1}{32} \sin(4x) + C, \quad C \in \mathbb{R}}$$

e. Rewrite the integral.

$$I = \int \frac{dx}{2x^2 - 3x + 2} = \int \frac{dx}{2x^2 - 3x + \frac{9}{8} + \frac{7}{8}} = \int \frac{dx}{\left(\sqrt{2}x - \frac{3\sqrt{2}}{4}\right)^2 + \frac{7}{8}}$$

$$= \frac{8}{7} \int \frac{dx}{\frac{8(\sqrt{2x} - \frac{3\sqrt{2}}{4})^2}{7} + 1}$$

Let $u = \frac{2\sqrt{2}}{\sqrt{7}} \left(\sqrt{2x} - \frac{3\sqrt{2}}{4} \right)$, then $du = \frac{4}{\sqrt{7}} dx$.

$$\begin{aligned} I &= \frac{8}{7} \int \frac{dx}{\frac{8(\sqrt{2x} - \frac{3\sqrt{2}}{4})^2}{7} + 1^2} = \frac{8}{7} \int \frac{1}{u^2 + 1} \cdot \frac{\sqrt{7}}{4} du = \frac{2}{\sqrt{7}} \int \frac{1}{u^2 + 1} du \\ &= \frac{2}{\sqrt{7}} \arctan u + C = \frac{2}{\sqrt{7}} \arctan \left[\frac{2\sqrt{2}}{\sqrt{7}} \left(\sqrt{2x} - \frac{3\sqrt{2}}{4} \right) \right] + C \\ &= \boxed{\frac{2}{\sqrt{7}} \arctan \left(\frac{4x - 3}{\sqrt{7}} \right) + C, \quad C \in \mathbb{R}} \end{aligned}$$

6. Use the method of integration by parts.

$$\begin{aligned} \left. \begin{aligned} u &= 1 + 2x \implies du = 2 dx \\ dv &= e^{-x} dx \implies v = -e^{-x} \end{aligned} \right\} \rightarrow \int u dv = uv - \int v du \\ \int_{-\infty}^0 (1 + 2x) e^{-x} dx &= \lim_{R \rightarrow -\infty} -e^{-x}(1 + 2x) \Big|_R^0 - \int_{-\infty}^0 -e^{-x} \cdot 2 dx \\ &= \lim_{R \rightarrow -\infty} (-e^0 \cdot 1 + e^{-R}(1 + 2 \cdot R)) + \lim_{P \rightarrow -\infty} 2e^{-x} \Big|_P^0 \\ &= -\infty + 2 \lim_{P \rightarrow -\infty} (e^0 - e^{-P}) = -\infty - \infty = \boxed{-\infty} \end{aligned}$$

The integral diverges to negative infinity.

7. The length of a curve defined by $x = f(y)$ whose derivative is continuous on the interval $a \leq y \leq b$ can be evaluated using the integral

$$S = \int_a^b \sqrt{1 + \left(\frac{dx}{dy} \right)^2} dy.$$

We have $\frac{dx}{dy} = \frac{2}{3} \cdot \frac{3}{2}(y-1)^{1/2} = \sqrt{y-1}$. Set $a = 1$, $b = 2$ and find the length.

$$S = \int_1^2 \sqrt{1 + \left(\sqrt{y-1} \right)^2} dy = \int_1^2 \sqrt{y} dy = \frac{2}{3} y^{3/2} \Big|_1^2 = \boxed{\frac{2}{3}(2\sqrt{2} - 1)}$$